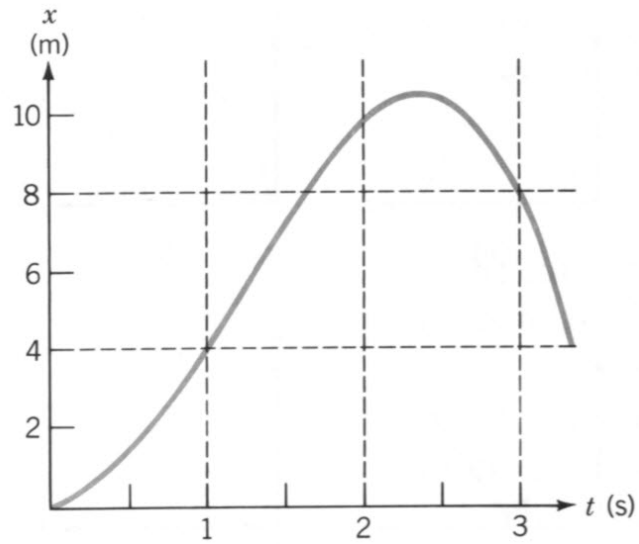
Homework #2

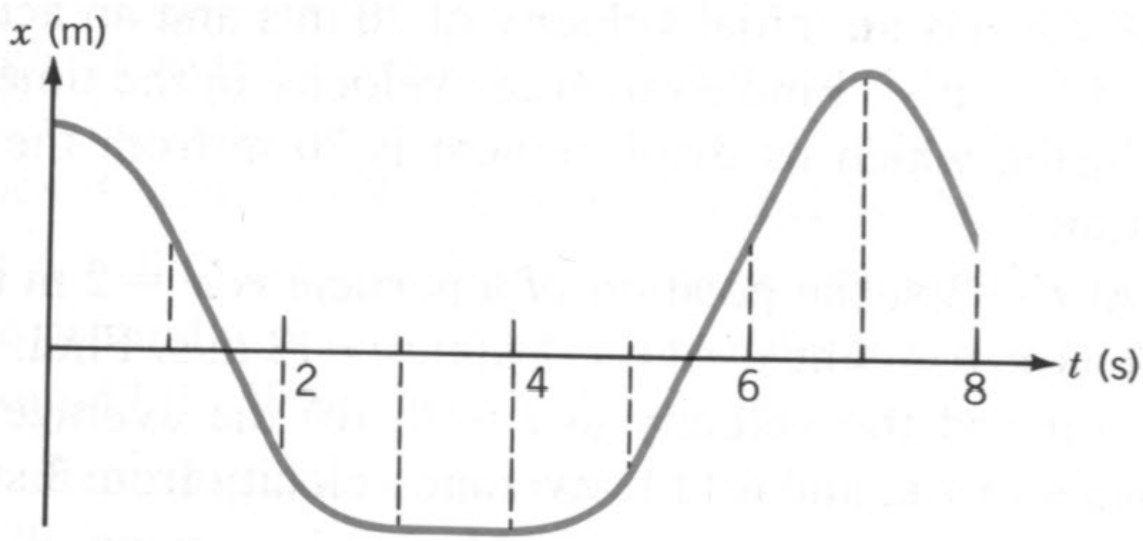
1) Using the graph estimate the instantaneous velocity at the following times: (a) 1.0 s; (b) 2.0 s; (c) 3.0 s.



2) A pitcher throws a baseball at 31.0 m/s. The batter hits it and gives it a velocity of 40.0 m/s in the opposite direction. If the ball and bat are in contact for 4.00×10^{-2} s, what is the average acceleration of the ball during this interval?

3) The position of a particle is given by $x = 4.00 - 5.00t + 3.00t^2$ (all constants and variables are in SI appropriate units) (a) What is its instantaneous velocity and acceleration at $t = 3.00$ s? (b) At what time is the particle at rest?

4) In the graph below, is there any time, or time interval, for which the following hold? (a) $v = 0$, $a = 0$; (b) $v = 0$, $a \neq 0$; (c) $v \neq 0$, $a = 0$; (d) $v > 0$, $a > 0$; (e) $v > 0$, $a < 0$; (f) $v < 0$, $a < 0$; (g) $v < 0$, $a > 0$.



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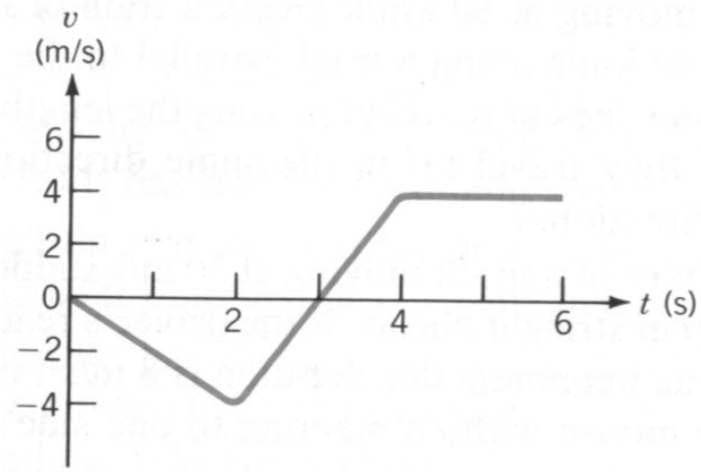
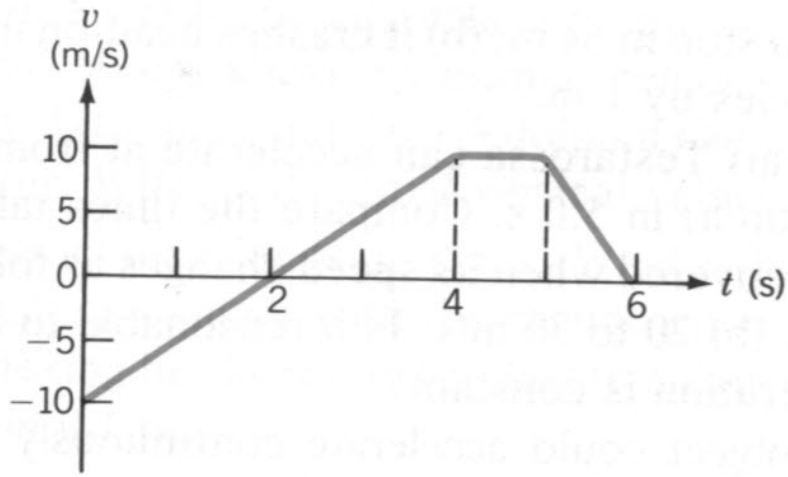
5) From the v versus t graph below estimate: (a) the average velocity for the first 6.0 s; (b) the average speed for the first 6.0 s.



6) From the v versus t graph below, plot the following graphs: (a) a versus t; (b) x versus t. (c) What is the average acceleration for the first 6.0 s? (d) What is the instantaneous acceleration at 2.0 s? Assume $x = 0$ at $t = 0$.



7) The length of a train is 44.5 m. Its front is 100. m from a pole. It accelerates from rest at 0.500 m/s^2. (a) How long does it take to go past the pole? (b) At what speeds do its front and rear pass the pole?

8) A bus slows with constant acceleration from 24.0 m/s to 16.0 m/s and moves 50.0 m in the process. (a) How much further does it travel before coming to a stop? (b) How long does it take to stop from 24.0 m/s?